Statistics Examples Sheet 1

This examples sheet covers material of the first 5 lectures and is appropriate for your first supervision. There will be two further examples sheets and a sheet of supplementary questions. A copy of this sheet can be found at: `http://www.statslab.cam.ac.uk/~rrw1/stats/`

1. (Lecture 1, unbiased estimation) Suppose X_1, X_2 are independent samples from $B(1, p)$. Let $T = X_1 + X_2$. In cases (a)–(c) show that $\hat{\theta}$ is an unbiased estimator of θ. Prove the statement made in case (d).

(a) $\theta = 2008 - p$, $\hat{\theta} = 2008 - \frac{1}{2}T$.

(b) $\theta = (1-p)^2$, $\hat{\theta} = 1$ if $T = 0$ and $\hat{\theta} = 0$ otherwise.

(c) $\theta = (1-3p)^2$, $\hat{\theta} = (-2)^T$.

(d) $\theta = \left(1 - \frac{1}{2}p\right)^{-1}$, there is no unbiased estimator of θ.

Hint: Note that $T \sim B(2, p)$ *and* $\mathbb{E}\hat{\theta}(T) = (1-p)^2\hat{\theta}(0) + 2p(1-p)\hat{\theta}(1) + p^2\hat{\theta}(2)$.

You should note from this example that an unbiased estimator can be silly (as in case (c) where $\hat{\theta} = -2$ when $T = 1$ even though we know $\theta > 0$), or may not even exist (as in case (d)).

2. (Lecture 2, MLE) In a genetics experiment, a sample of n individuals was found to include a, b, c of the three possible genotypes GG, Gg, gg respectively. The population frequency of a gene of type G is $\theta/(\theta + 1)$, where θ is unknown, and it is assumed that the individuals are unrelated and that two genes in a single individual are independent. Show that the likelihood of θ is proportional to

$$\theta^{2a+b} / (1+\theta)^{2a+2b+2c}$$

and that the maximum likelihood estimate of θ is $(2a + b)/(b + 2c)$.

3. (Lecture 2, MLE and sufficiency) Suppose $X_1, \ldots, X_n$ is a random sample from a gamma (α, λ)distribution with density function

$$f(x \mid \alpha, \lambda) = \frac{\lambda^\alpha x^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)}, \quad x > 0.$$

Let $\theta = (\alpha, \lambda)$. What is meant by saying that $T(X)$ is sufficient for θ? Find a sufficient statistic for θ. How might you find MLEs for α and λ?

Hint. In this example the sufficient statistic is a vector with two components.

4. (Lecture 2, MLE and sufficiency) In each of cases (a)–(c) write down the likelihood of θ and show that the stated $T(X)$ is a sufficient statistic for θ.

In each case also find a MLE of θ and show that it is a function of $T(X)$. Find the distribution of $T(X)$ and determine whether or not the MLE is an unbiased estimator of θ. If it is not, verify that it is asymptotically unbiased, and find some other estimator which is unbiased.

(a) $X_1, \ldots, X_n$ are independent Poisson random variables, with X_i having mean $i\theta$, where $\theta > 0$. $T(X) = \sum_{i=1}^n X_i$.

(b) $X_1, \ldots, X_n$ are independent normal random variables, with $X_i \sim N(\theta, \sigma_i^2)$ and σ_i^2, $i = 1, \ldots, n$, known. $T(X) = \sum_{i=1}^n X_i/\sigma_i^2$.

(c) $X_1, \ldots, X_n$ are $n > 2$ independent and exponentially distributed random variables, with parameter θ, i.e., with density $f(x \mid \theta) = \theta e^{-\theta x}$, $x > 0$. $T(X) = \sum_{i=1}^n X_i$.

Hint: In case (a), $T(X) \sim P\left(\frac{1}{2}n(n+1)\theta\right)$. *In case (b),* $T(X) \sim N\left(\theta \sum_i \sigma_i^{-2}, \sum_i \sigma_i^{-2}\right)$. *In case (c),* $T(X) \sim gamma(n, \theta)$. *Do you understand why?*

5. (Lecture 3, Rao-Blackwell theorem) Suppose $X_1, \ldots, X_n$ are independent random variables with distribution $B(1, p)$.

(a) Show that a sufficient statistic for $\theta = (1-p)^2$ is $T(X) = \sum_{i=1}^n X_i$ and that the MLE for θ is $\left(1 - \frac{1}{n}T\right)^2$.
Hint: Use the chain rule, $df/d\theta = (df/dp)(dp/d\theta)$.

(b) The MLE is a biased estimator for θ. Find a function of T which is an unbiased estimator for θ.
Hint: $\theta = \mathbb{P}(X_1 + X_2 = 0)$. *Recall example 1(b) above.*

6. (Lecture 3, Rao-Blackwell theorem) Suppose $X_1, \ldots, X_n$ are independent random variables uniformly distributed over $(\theta, 2\theta)$. Show that a sufficient statistic for θ is $T(X) = (\min_i X_i, \max_i X_i)$ and that an unbiased estimator of θ is $\hat{\theta} = \frac{2}{3}X_1$. Find an unbiased estimator of θ which is a function of $T(X)$ and whose mean square error is no more than that of $\hat{\theta}$.

Note that this is another example in which the sufficient statistic turns out to be a vector, despite the fact that the parameter θ is only a scalar.

7. (Lecture 4, confidence intervals) A random variable is uniformly distributed over $(0, \theta)$. Show that the maximum of a random sample of n values of this variable is sufficient for θ and that this is also the MLE for θ. Show also that a $100\gamma\%$ confidence interval for θ is $(y_n, y_n/(1-\gamma)^{1/n})$, y_n being the maximum of the sample.

8. (Lecture 4, confidence intervals) Suppose that $X_1 \sim N(\theta_1, 1)$ and $X_2 \sim N(\theta_2, 1)$ independently, where θ_1 and θ_2 are unknown. For this model, $(\theta_1 - X_1)^2 + (\theta_2 - X_2)^2$ has the distribution $\mathcal{E}(\frac{1}{2})$, i.e., the exponential distribution with mean 2. (A fact you may recall from Probability IA, and which we will prove again later.)

Show that both the square S and circle C in $\mathbb{R}^2$, given by

$$S = \{(\theta_1, \theta_2) : |\theta_1 - X_1| \leq 2.236;\ |\theta_2 - X_2| \leq 2.236\}$$
$$C = \big\{(\theta_1, \theta_2) : (\theta_1 - X_1)^2 + (\theta_2 - X_2)^2 \leq 5.991\big\}$$

are 95% confidence regions for (θ_1, θ_2), in the sense that $\mathbb{P}(S \text{ contains } (\theta_1, \theta_2)) = 0.95$ and $\mathbb{P}(C \text{ contains } (\theta_1, \theta_2)) = 0.95$. *Hint:* $\Phi(2.236) = (1 + \sqrt{.95})/2$, *where* Φ *is the cdf of* $N(0, 1)$.

Which of S and C would you prefer, and why?

9. (Lecture 5, Bayes estimation) Each word that baby Hamlet speaks is chosen independently and with equal probability from a set of k words. Suppose your prior belief is that k is equally likely to be either 5, 6, 7 or 8. You hear him say 'to not be or be to'. Show that the posterior probability mass function of k is proportional to $q(k) := (k-1)(k-2)(k-3)/k^5$, $k = 5, 6, 7, 8$, and is 0 otherwise.

Given that $q(k)$ has values 0.00768, 0.00772, 0.00714, 0.00641 for $k = 5, 6, 7, 8$ respectively, find a point estimate of k under the loss function

$$L(k, \hat{k}) = \begin{cases} 0 & \text{if } \hat{k} = k, \\ 1 & \text{if } \hat{k} \neq k. \end{cases}$$

How does this particular choice of prior distribution and loss function relate to maximum likelihood estimation?

10. (Lecture 5, Bayes estimation) Suppose that the number of defects on a roll of magnetic recording tape has a Poisson distribution for which the mean λ is known to be either 1 or 1.5. Suppose the prior mass function for λ is

$$\pi_\lambda(1) = 0.4, \quad \pi_\lambda(1.5) = 0.6\,.$$

A collection of 5 rolls of tape are found to have $x = (3, 1, 4, 6, 2)$ defects respectively. Show that the posterior distribution for λ is

$$\pi_\lambda(1 \mid x) = 0.012, \quad \pi_\lambda(1.5 \mid x) = 0.988\,.$$

You will have to use your calculator for this one.

11. (Lecture 5, Bayes estimation) Suppose $X_1, \ldots, X_n$ are IID from a distribution uniform on $(\theta - \frac{1}{2}, \theta + \frac{1}{2})$, and that the prior for θ is uniform on (10,20). Calculate the posterior distribution for θ, given $x = X_1, \ldots, X_n$ and show that the point estimate for θ under both quadratic and absolute error loss functions is

$$\hat{\theta} = \tfrac{1}{2}\left[\max_i(x_i - \tfrac{1}{2}) \vee 10 + \min_i(x_i + \tfrac{1}{2}) \wedge 20\right].$$

The notation here is $a \vee b = \max\{a, b\}$ and $a \wedge b = \min\{a, b\}$.

12. (Lecture 5, Bayes estimation) Suppose $X_1, \ldots, X_n$ form a random sample from the following pdf:

$$f(x \mid \theta) = \begin{cases} \theta x^{\theta - 1} & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

and that the prior for θ is gamma(α, β), $\alpha > 0, \beta > 0$, with density

$$\pi(\theta) = \frac{\beta^\alpha \theta^{\alpha - 1} e^{-\beta\theta}}{\Gamma(\alpha)}, \quad \theta > 0\,.$$

Show that the posterior distribution of θ is gamma $(\alpha + n, \beta - \sum_i \log x_i)$ and hence that a point estimate for θ under quadratic loss function is

$$\frac{\alpha + n}{\beta - \sum_{i=1}^n \log x_i}.$$

Hint: You may want to refer to the notes for Lecture 1 to remind yourself of some basic facts about the gamma distribution.

13. (Lecture 5, Bayes estimation) Suppose that X is distributed as a binomial random variable $B(n, \theta)$. Suppose the prior distribution for θ is the uniform distribution on $[0, 1]$ and the loss function is

$$L(\theta, \hat{\theta}) = (\theta - \hat{\theta})^2 / \theta(1 - \theta)\,.$$

Show that, based on the single observation x, the point estimate for θ is $\hat{\theta} = x/n$.

Hint: You may want to refer to the notes for Lecture 1 to remind yourself of some basic facts about the beta distribution. Recall

$$\int_0^1 x^{a-1}(1 - x)^{b-1}\, dx = B(a, b) := \frac{\Gamma(a)\Gamma(b)}{\Gamma(a + b)}$$

and that $\Gamma(a) = (a - 1)!$ *when* a *is an integer.*

R Weber, January 22, 2008

Statistics Examples Sheet 2

This examples sheet covers material of lectures 6–10 and is appropriate for your second supervision. A copy of this sheet can be found at: `http://www.statslab.cam.ac.uk/~rrw1/stats/`

1. (Lecture 6, Hypothesis testing) A positive random variable X has density function

$$f(x \mid \theta) = \frac{\theta}{(\theta + x)^2}, \quad x > 0$$

where $\theta > 0$ is an unknown parameter. Find the best test of size 0.05 of $H_0 : \theta = 1$ vs $H_1 : \theta = 2$, and show that the probability of Type II error is 19/21.

2. (Lecture 6, Hypothesis testing) Let $X_1, X_2, \dots, X_n$ be IID random variables, each with the Poisson distribution of parameter θ (and therefore of mean θ and variance θ). Find the best size α test of $H_0 : \theta = 1$ against $H_1 : \theta = 1.21$. By using the Central Limit Theorem to approximate the distribution of $\sum_i X_i$ show that the smallest value of n required to make $\alpha = 0.05$ and $\beta \leq 0.1$ (where α and β are the Type I and Type II error probabilities) is somewhere near 213.

3. (Lecture 7, Further hypothesis testing) Find the best size α test of $H_0 : \theta = \theta_0$ vs $H_1 : \theta = \theta_1$ $(> \theta_0)$ and write down an expression for the power function when $X_1, \dots, X_n$ are IID exponential random variables, with parameter θ.

Use χ^2 tables to find the least sample size which will allow us to test $H_0 : \theta = 1$ against $H_1 : \theta = 3$ with $\alpha = 5\%$ and $\beta \leq 10\%$. Describe the appropriate critical region numerically. [*Hint: Recall the equivalence of the gamma*$(n/2, 1/2)$ *and* χ^2_n *distributions.*]

4. (Lecture 7, Further hypothesis testing) Let $X_1, X_2, \dots, X_n$ be a sample of size n from the distribution with probability density function

$$\frac{1}{2}\lambda e^{-\lambda|x|}, \quad -\infty < x < \infty, \quad \lambda > 0,$$

where λ is unknown. It is desired to test the hypothesis $H_0 : \lambda = \lambda_0$ against $H_1 : \lambda = \lambda_1$ where $\lambda_1 > \lambda_0$. Find the test that minimizes the sum of the probabilities of the two types of error. Denote the size of this test by α. Is this the most powerful test of size α for testing $H_0 : \lambda = \lambda_0$ against $H_1 : \lambda = \lambda_2$, for each $\lambda_2 > \lambda_0$?

5. (Lecture 8, Generalized likelihood ratio tests) Let $X_1, \dots, X_n$ be an IID random sample from an exponential distribution $\mathcal{E}(\theta_1)$, with density

$$f(x \mid \theta_1) = \theta_1 e^{-\theta_1 x}, \quad x > 0$$

and let $Y_1, \dots, Y_n$ be an independent IID random sample from $\mathcal{E}(\theta_2)$.

Find the form of the likelihood ratio test for testing $H_0 : \theta_1 = \theta_2$ against $H_1 : \theta_1 \neq \theta_2$. Show that the test can be expressed in terms of the statistic

$$T = \textstyle\sum_i X_i / (\sum_i X_i + \sum_i Y_i).$$

By showing that when H_0 is true the distribution of T does not depend on $\theta = \theta_1 = \theta_2$, construct a test of exact size α of H_0 against H_1 based on T.

6. (Lecture 8, Generalized likelihood ratio tests) The data $x_1, \dots, x_n$ has been observed and it is known that x_i is a sample from a Poisson distribution with an unknown mean λ_i. It is desired to test $H_0 : \lambda_1 = \cdots = \lambda_n$ against a general alternative hypothesis that the λ_i are arbitrary. Show that, on the basis of the generalized likelihood ratio test, H_0 should be rejected for large values of the test statistic

$$2 \sum_{i=1}^{n} x_i \log(x_i / \bar{x}),$$

where $\bar{x} = (1/n) \sum_i x_i$. Show that this statistic is approximately

$$\frac{1}{\bar{x}} \sum_{i=1}^{n} (x_i - \bar{x})^2.$$

What would you conclude for data $(3, 5, 1, 6, 5)$?

7. (Lecture 9, Chi-squared tests of categorical data)

(i) It is known that an observation made on a certain system will yield a result falling into one of k categories, $C_1, C_2, \dots, C_k$. To each value of the real parameter θ in a given interval corresponds a probability distribution $p_i(\theta)$ $(i = 1, 2, \dots, k)$. The null hypothesis is made that, for some unknown value of θ, the probability of a result falling into category C_i is $p_i(\theta)$. A total of n observations is made, of which n_i fall into category C_i $(i = 1, 2, \dots, k)$. Show carefully how to use a χ^2 distribution in testing the above hypothesis, and describe how you would carry out the test, using a statistic of the form

$$\sum_{i=1}^{k} (n_i - e_i)^2 / e_i,$$

where e_i is the expected number of observations in category C_i under (an appropriate case of) the null hypothesis.

(ii) A scientist gets observations in three categories, across which he suspects a linear trend of probability, in which case

$$p_1(\theta) = \frac{1}{3} - \theta, \quad p_2(\theta) = \frac{1}{3}, \quad p_3(\theta) = \frac{1}{3} + \theta$$

for some value of θ such that $-\frac{1}{3} < \theta < \frac{1}{3}$. The numbers observed are $n_1 = 12$, $n_2 = 24$, $n_3 = 24$. Test the hypothesis of linearity.

8. (Lecture 9, Chi-squared tests of categorical data) A machine produces plastic articles in bunches of three articles at a time. The process is rather unreliable, and quite a few defective articles are observed. In an experimental run of the machine, 512 bunches were produced. Of these, the numbers of bunches with $i = 0, 1, 2, 3$ defective articles were 213 ($i = 0$), 228 ($i = 1$), 57 ($i = 2$), and 14 ($i = 3$). Test the hypothesis that each article has a constant (but unknown) probability θ of being defective, independently of all other articles.

9. (Lecture 9, Chi-squared tests of categorical data) From each of six batches of seed a random sample of 100 seeds was selected for sowing. The numbers of seeds that failed to germinate in the six samples of 100 seeds were

$$12, 20, 9, 17, 24, 16.$$

Test the hypothesis that the proportion of non-germinating seeds was the same for all batches.

10. (Lecture 9, Chi-squared tests of categorical data) The following data is for clinical trials of old and new treatments for a disease. Exactly 1,100 patients were chosen to receive each treatment.

	Survive	Die	Total
Old	505	595	1,100
New	195	905	1,100
Total	700	1,500	2,200

Test the hypotheis that the old and new treatments are equally successful. On this basis, which treatment would you prefer?

In fact, the trials took place at two hospitals, for which the data is given below. Doctors at Hospital A, a famous research hospital, designed the trial. Their patients tend to be more seriously ill and they also used the new treatment more often. Do you wish to revise your conclusion about the relative effectiveness of the two treatments?

	Hospital A		
	Survive	Die	Total
Old	5	95	100
New	100	900	1,000
Total	105	995	1,100

	Hospital B		
	Survive	Die	Total
Old	500	500	1,000
New	95	5	100
Total	595	505	1,100

What lesson do you learn from this example?

11. (Lecture 9, Chi-squared tests of categorical data) A random sample of 59 people from the planet Krypton yielded the following results:

		Eye-colour	
		1 (Blue)	2 (Brown)
Sex	1 (Male)	19	10
	2 (Female)	9	21

Professor A had believed that sex and eye-colour are independent factors on Krypton. After doing a χ^2 test of his hypothesis against the alternative

$$H_1 : \sum_i \sum_j p_{ij} = 1,$$

he finds that he has to reject it at the 5% level and even at the 1% level.

Professor B has always believed the much stronger hypothesis that $p_{ij} = \frac{1}{4}$ for all i and j. After doing a χ^2 test of his hypothesis against H_1, he finds that he does not need to reject it at the 5% level.

You are asked for expert statistical advice: firstly, to check the calculations, and secondly, to comment on whether or not Statistics is an absurd subject.

12. (Lecture 9, Chi-squared tests of categorical data) In Exercises 10 and 11 you have performed Pearson χ^2 tests. Explain carefully how the form of the null and alternative hypotheses in these exercises differ.

13. (Lecture 10, Distributions of the sample mean and variance) Statisticians A and B obtain independent IID samples, $X_1, \ldots, X_{10}$ and $Y_1, \ldots, Y_{17}$ respectively, from a $N(\mu, \sigma^2)$ distribution, for which μ and σ^2 are both unknown. They estimate (μ, σ^2) by $(\bar{X}, S_{XX}/9)$ and $(\bar{Y}, S_{YY}/16)$, respectively. Given that $\bar{X} = 5.5$ and $\bar{Y} = 5.8$, which statistician's estimate of σ^2 is more probable to have exceeded the true value of σ^2 by more than 50%? Find this probability (approximately) in each case.

Hint: This is something of a 'trick' question. Why? You may find χ^2 tables helpful.

R Weber, January 22, 2008

Statistics Examples Sheet 3

This examples sheet covers material of the lectures 11–16 and is appropriate for your third supervision. A copy of this sheet can be found at: `http://www.statslab.cam.ac.uk/~rrw1/stats/`

On this examples sheet *numeric answers* are required and you will also need to consult statistical tables. You can use a pocket calculator. A computer spreadsheet is also a good way to do the calculations.

1. (Lecture 11, The t-test) A sample of nine pieces of steel wire of type A yields the following breaking strengths: 11.99, 12.02, 12.03, 12.09, 12.14, 12.16, 12.16, 12.16, 12.23. Use a t-test to test for any difference in mean breaking strength between wire A and a standard type whose mean breaking strength is known to be 12.10 units.

2. (Lecture 11, The t-test) The following figures give the % extension under a given load for two random samples of lengths of yarn, the first sample being taken before washing and the second after six washings.

Before washing:	12.3	13.7	10.4	11.4	14.9	12.6	
After 6 washings:	15.7	10.3	12.6	14.5	12.6	13.8	11.9

Do they provide any justification for concluding that extensibility is affected by washing? (You may assume that the standard deviation is unaltered by the washings.)

3. (Lecture 11, The t-test) In another experiment with the same type of yarn, six lengths of yarn were selected at random and each length cut into two halves. One of the halves was tested for extension without washing, and the other after six washings, giving the following % extensions:

Length:	A	B	C	D	E	F
Before washing:	13.9	12.5	11.0	11.8	10.8	14.6
After 6 washings:	14.7	12.1	13.2	13.6	11.5	15.4

What evidence do these provide as to the effect of washing on extensibility? Why should this experiment be analysed differently from the previous one?

4. (Lecture 11, The t-test) The following data is sampled from a $N(\mu, \sigma^2)$ distribution where both μ and σ^2 are unknown.

$$6.82 \quad 6.07 \quad 3.74 \quad 6.87 \quad 5.92$$

For this data, $\sum_i x_i = 29.42$ and $\sum_i x_i^2 = 179.588$.

(a) Find a 95% confidence interval for μ and show that its width is about 3.16.

(b) Suppose it becomes known that the true value of σ^2 is 1. Show that the 95% confidence interval for μ now has a width of 1.76.
Note that the width of the confidence interval has turned out to be narrower when the true value of σ^2 is known. Will this always happen?

(c) Consider the event that the 95% confidence interval for μ is narrower when σ^2 is known than it is when σ^2 is unknown, given that both intervals are computed from the same data, $X_1, \dots, X_n$, where these are IID samples from $N(\mu, \sigma^2)$.
Show that if $n = 5$ then this event has probability $1 - F_4(a)$, where $a = 4(1.96/2.78)^2 = 1.988$, and F_4 is the cdf of the χ^2_4 distribution. Verify from tables that this probability is a bit less than 0.75 (actually, 0.737).

5. (Lecture 12, The F-test and analysis of variance)

A butter-packing machine in a dairy packs butter in 250g packages though the actual weights packed vary. In a trial run, the following deviations from the nominal 250g were recorded:

$$5, 8, 0, 3, -1, 1, 6, 5, 8, 4, 9, 0, 4.$$

It is proposed to replace the machine with a new model which, it is claimed, has smaller variability in package weights. A trial run on the new machine produced the following deviations from the nominal 250g:

$$3, -2, 1, 4, 0, 2, 1, 5, 3, 2, 1, 4.$$

Test a null hypothesis that the variability of new machine is no better than the old.

Suppose the variance of the new machine is actually one-third that of the old, i.e. $\sigma^2_{\text{new}} = \frac{1}{3}\sigma^2_{\text{old}}$. On the basis of 13 trials of the old machine and 12 trials of the new machine a 5% level test is to be made of $H_0 : \sigma^2_{\text{new}} = \sigma^2_{\text{old}}$ against $H_1 : \sigma^2_{\text{new}} < \sigma^2_{\text{old}}$. What is the probability that a type II error will be made? *Hint:* $F^{(12,11)}_{0.55} = 0.932$.

6. (Lecture 12, The F-test and analysis of variance) An experiment was carried out to determine whether four specific temperatures used in the firing of bricks affected the density of the brick. The experiment led to the following data for 5 samples at 110° and 120°, and 4 at 130° and 140°:

°C	110°	120°	130°	140°
data	20.8	20.6	20.9	20.8
	20.9	20.3	20.8	20.9
	20.7	20.2	20.8	20.9
	20.6	20.3	20.7	20.4
	20.7	20.5		

Carry out a one-way ANOVA to test for an effect of temperature upon density.

7. (Lecture 13, Linear regression and least squares) Show that the least squares estimator of β in a model $Y_i = \beta x_i + \epsilon_i$, (a regression through the origin), is $\hat{\beta} = \sum_i x_i Y_i / \sum_i x_i^2$. Under what assumptions is this also the MLE?

The range (in m) of a howitzer with muzzle velocity $V\ m/s$ fired at angle of elevation α is $g^{-1}V^2 \sin 2\alpha$, where $g = 9.81$. Suppose that 9 shells are fired at different angles of elevation, with the following results:

α	5	10	15	20	25	30	35	40	45
$\sin 2\alpha$	0.1736	0.3420	0.5	0.6428	0.7660	0.8660	0.9397	0.9848	1
range	4,860	9,580	14,080	18,100	21,550	24,350	26,400	27,700	28,300

(Ranges are given in m, and angles in degrees.) Estimate V from these data.

8. (Lecture 13, Linear regression and least squares) A beaker of liquid is heated, and then removed from the source of heat and left to cool in room whose temperature is 9°C. The temperature of the liquid is measured at the moment it is removed from the heat, and at intervals of 1 minute thereafter. The following results are observed:

time (secs)	0	1	2	3	4	5	6
temperature (°C)	100	82	60	50	40	32	28

Set up, and study, an appropriate regression model. *Hint: The temperature of the liquid decays towards room temperature exponentially with time.*

9. (Lecture 14, Hypothesis tests in regression models) Carry out a linear regression analysis on the following data:

x	-3	-3	-2	-1	0	1	1	2	2	3
y	114	110	111	107	107	108	104	105	103	101

Find a 95% confidence interval for β, the true slope of the regression line, and for the mean value of Y when $x = 1$.

10. (Lecture 14, Hypothesis tests in regression models) Five groups of 25 male fruit-flies have been kept under quite different conditions. Within each group longevity (y) is positively correlated with thorax size (x). Suppose that in the ith group

$$y_{ij} = \alpha_i + \beta_i x_{ij} + \epsilon_{ij}, \quad j = 1, \dots, 25,$$

where ϵ_{ij} are IID $N(0, \sigma^2)$. It is desired to test $H_0 : \beta_1 = \cdots = \beta_5$ against the alternative that H_0 is not true. On minimizing $S = \sum_{i=1}^{5} \sum_{j=1}^{25} (y_{ij} - \alpha_i - \beta_i x_{ij})^2$, with respect to $\alpha_1, \dots, \alpha_5$, $\beta_1, \dots, \beta_5$ under constraints appropriate to H_0 and H_1, we find minimized values of $R_0 = 12821.0$ and $R_1 = 12633.2$ respectively. Test H_0, saying how you determine the degrees of freedom used in your test. You may assume, by analogy with similar tests you have seen, appropriate distributional results for R_0, R_1 and $R_0 - R_1$. (The calculation required to answer this question is trivial. The important thing is to see if you can understand the rationale that underlies it.)

11. (Lecture 15, Computational methods) Twelve language students were asked to read an article in Japanese and then answer 25 multiple-choice questions about the article. Six students (randomly chosen) were presented with the questions written in English and six were presented with the questions in Japanese. Their scores were

Questions written in English:	12	17	19	22	25	25
Questions written in Japanese:	10	11	13	16	21	24

It is desired to test a null hypothesis that the language in which the questions are presented makes no difference to the students' performances. To do this a computer program was used to randomly sample 6 numbers without replacement from the set of 12 numbers above and to compute their total. This process was repeated 200 times. Totals of 85, 87, 88, 89, 90, ..., 120, 121, 122, 123 and 130 were observed to occur 1, 2, 1, 3, 5,..., 5, 8, 3, 1 and 1 times respectively.

Use this information to conduct a 5% level test of the null hypothesis, explaining the thinking that underlies your test. What do you think are the advantages and disadvantages of this test (which is known as a 'permutation test') compared to the two-sample t-test for equality of means?

12. (Lecture 16, Decision theory and Bayesian inference) A random variable X is observed. If it comes from class A its distribution is $N(0,1)$, and if it comes from class B its distribution is $N(1,25)$. Classes A and B are equally likely, so the prior is $\mathbb{P}(A) = \mathbb{P}(B) = \frac{1}{2}$. Under 0–1 loss, what is the Bayes rule for classification (i.e., what rule minimizes the probability of misclassification)? How should $X = 1.5$ be classified? What is the probability that when $X = 1.5$ a misclassification occurs?

R Weber, January 22, 2008

Statistics Further Examples Sheet

This examples sheet has some extra questions which you may like to do for further practice or revision. A copy of this sheet can be found at: `http://www.statslab.cam.ac.uk/~rrw1/stats/`

1. (Sufficiency) Let $X_1, \dots, X_n$ be independent Poisson random variables with X_j having parameter $j\lambda$, where $\lambda > 0$ is an unknown parameter. Find a sufficient statistic for λ. What is its distribution?

2. (Sufficiency, MLE, confidence intervals) Suppose that $X_1, \dots, X_n$ are independent random variables, $X_j \sim N(\theta, \sigma_j^2)$, where θ is an unknown parameter, but the σ_j^2, $j = 1, \dots, n$ are known. Find a sufficient statistic for θ. What is its distribution? Find the MLE of θ, and find a 95% confidence interval for θ.

3. (Sufficiency, MLE, confidence intervals) Suppose that $n > 2$, and

$$f(x \mid \lambda) = \begin{cases} \lambda e^{-\lambda x} & (x \geq 0), \\ 0 & (x < 0). \end{cases}$$

Find a sufficient statistic $T_n = T_n(X_1, \dots, X_n)$ for λ, and write down its density. Obtain the maximum-likelihood estimator for λ and show that it is biased, but that some multiple of it is not. Find the distribution of $U_n = 2\lambda \sum_{i=1}^n X_i$ and hence obtain a 95% confidence interval for λ.

4. (Rao-Blackwellization) The random variables $X_1, \dots, X_n$ are IID Poisson with mean μ. Write down a simple unbiased estimator of $\theta = P(X_1 = 0)$. Is it a function of the sufficient statistic for μ? If not, find a better estimate by conditioning on the sufficient statistic.

5. (Bayes estimation) Show that the posterior mean can never be an unbiased estimator of θ.

Hint: Show that $\mathbb{E}_\theta[E_X(X-\theta)^2]$ *and* $\mathbb{E}_X[E_\theta(X-\theta)^2]$ *are respectively* $\mathbb{E}\theta^2 - \mathbb{E}X^2$ *and* $\mathbb{E}X^2 - \mathbb{E}\theta^2$ *and hence that* $\mathbb{E}(X-\theta)^2 = 0$.

6. (Lecture 9. Chi-squared tests of categorical data) Show that in a 2×2 contingency table, with observed frequencies Y_{ij} $(i, j = 1, 2)$, the usual expression for the statistic X^2 may be put in the form

$$Y_{++}(Y_{11}Y_{22} - Y_{12}Y_{21})^2 / (Y_{1+}Y_{2+}Y_{+1}Y_{+2})$$

where $+$ denotes summation over both values of the corresponding subscript.

7. (regression) Suppose that for given $x_1, \dots, x_n$,

$$Y_i \sim N(\beta x_i, x_i \sigma^2), \quad i = 1, \dots, n,$$

independently. Show that this is equivalent to the model

$$\frac{Y_i}{\sqrt{x_i}} = \beta\sqrt{x_i} + \epsilon_i,$$

where $\epsilon_1, \dots, \epsilon_n$ are IID $N(0, \sigma^2)$. Show that both the MLE and LSE of β is $\hat{\beta} = \bar{Y}/\bar{x}$.

A nationwide chain of 500 retail shops has just started its annual sale. Management would like to estimate, by 9pm on the first day of the sale, the total sales there have been that first day. It is difficult to collect data from all shops by 9pm, but this can be done for 50 of the shops. The shops differ in size: shop i is of size x_i; first day sales in shop i are y_i. The total size of shops in the chain, $\sum_{i=1}^{500} x_i$, is known, as are the pairs (x_i, y_i), for the sample $i = 1, \dots, 50$.

Explain, using the ideas in the regression model at the start of the question, how you could estimate the total sales. What assumption are you making about the relationship between sales and shop size? Is this reasonable? What assumption are you making about the sample of 50 shops? Supposing that all these assumptions are valid, show that your estimator of $\sum_{i=1}^{500} y_i$ is unbiased.

8. (regression) Suppose that n points are to be chosen in the interval $[-1, 1]$ for estimating a and β in the regression model

$$y_i = a + \beta x_i + \epsilon_i,$$

where the $\epsilon_1, \dots, \epsilon_n$ are IID $N(0, \sigma^2)$. What values should be given to $x_1, \dots, x_n$ so as to minimize $\mathrm{var}(\hat{\beta})$?

9. (Lecture 4. confidence intervals) Suppose X_1, X_2 two IID samples from a uniform distribution on $\left(\theta - \frac{1}{2}, \theta + \frac{1}{2}\right)$. Show that $(\min_i x_i, \max_i x_i)$ is a 50% confidence interval for θ. Suppose you observe the data $x = (4.0, 3.4)$. How certain are you that θ lies in the interval $(3.4, 4.0)$?

Hint: $\mathbb{P}(\min_i x_i \leq \theta \leq \max_i x_i) = \mathbb{P}(X_1 \leq \theta \leq X_2) + \mathbb{P}(X_2 \leq \theta \leq X_1)$. *Why?*

10. (contingency table)

11. (theory of multivariate normal) Define the *multivariate normal* distribution as follows. If $\mathbf{y}$ is a p-vector with p-dimensional pdf:

$$f_{\mathbf{y}}(\mathbf{y}) = \frac{1}{|2\pi\Sigma|^{\frac{1}{2}}} \exp -\frac{1}{2}\{(\mathbf{y}-\mu)^T\Sigma^{-1}(\mathbf{y}-\mu)\}, \mathbf{y} \in \mathbb{R}^p,$$

where μ is a p-vector and Σ a symmetric, positive definite $p \times p$ matrix, then $\mathbf{Y}$ is said to have a multivariate normal distribution, and we write

$$\mathbf{Y} \sim \mathrm{N}_p(\mu, \Sigma).$$

Prove the following:

(i) $Y_1, \ldots, Y_p$ are independent if and only if Σ is diagonal. Also show that when $\Sigma = kI$, $\mu = \mu\mathbf{1}$ then $Y_1, \ldots, Y_p$ are IID.

(ii) If $\mathbf{Y} \sim \mathrm{N}_p(\mu, \Sigma)$, then $\mathbf{Z} = A\mathbf{Y} \sim \mathrm{N}_p(A\mu, A\Sigma A^T)$. (Note the special case: Lemma 3.4.1 from notes.)

(iii) Show that $\Sigma^{-\frac{1}{2}}(\mathbf{Y}-\mu) \sim \mathrm{N}_p(0, I)$.

(iv) From (iii), deduce that $\mathbb{E}[\mathbf{Y}] = \mu$, and that the ij^{th} entry of Σ, σ_{ij}, is $\mathrm{cov}(Y_i, Y_j)$.

12. (theory of bivariate normal) Suppose (X_i, Y_i) $1 \le i \le n$ are IID from a bivariate normal population with mean $\mu = (\mu_X, \mu_Y)^T$, and $\Sigma = \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix}$.

(a) Calculate the conditional distribution of Y_i given X_i.

(b) Suppose we attempt to fit the linear model

$$Y_i = \alpha + \beta(x_i - \bar{x}) + \epsilon_i.$$

From (a), argue that the hypotheses $H_0 : \beta = 0$ and $H_0 : \rho = 0$ are equivalent.

(c) Letting $S^2 = (n-2)^{-1}\sum_{i=1}^n (Y_i - \hat{\alpha} - \hat{\beta}(x_i - \bar{x}))^2$, and $S_{XX} = \sum_{i=1}^n (x_i - \bar{x})^2$, show that

$$\frac{\sqrt{S_{XX}}\hat{\beta}}{S} = \sqrt{n-2}\frac{r}{\sqrt{1-r^2}}.$$

where r is the sample covariance, $\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})/(\Sigma(y_i - \bar{y})^2\Sigma(y_i - \bar{x})^2)^{\frac{1}{2}}$. Therefore construct a test of $H_0 : \beta = 0$ depending only on r^2 and n.

13. (Strong Likelihood Principle) On the Planet Altair A, all nuclear power stations were installed exactly 20 years ago by ANFL. It is known that cataclysmic accidents occur at a Poisson process of constant (but unknown) rate λ per year. The first cataclysmic accident has just happened. ANFL argue that it is not fair to apply statistics just after an accident has occurred. However GreenAlt, the local conservation movement, argues as follows:

1°. The likelihood of getting precisely 1 accident (some time) in a 20-year period is

$$(20\lambda)e^{-20\lambda}$$

by the formula for the Poisson distribution.

2°. The likelihood that the first accident occurs at time 20 is

$$\lambda e^{-20\lambda}$$

by the formula for the exponential distribution.

3°. Since these likelihoods are proportional and since the constant of proportionality, 20, will drop out of every statistical consideration, any inferences about λ made on the basis that the first accident has just occurred must be the same as those based on the fact that precisely 1 accident has occurred in all in 20 years.

Most of the people-in-the-street on Altair A believe that there is substance in ANFL's argument and that GreenAlt are trying to overturn common sense with silly mathematics. What do you think? The Problem is not unknown on Planet Earth; and GreenAlt's argument, if valid, remains valid if 1 accident is replaced by 3 or 4.